

SAR Based Flood Inundation and Agricultural Impact Assessment

Karamchedu Mandal, Bapatla District, Andhra Pradesh

1. Introduction

Flood inundation caused by tropical cyclones poses a significant threat to agriculture, infrastructure, and livelihoods in coastal and deltaic regions of India. Rapid and reliable assessment of flood extent and its impact on vegetation is essential for disaster response and damage estimation.

This study presents an **EO-SAR based flood inundation and agricultural impact assessment** for **Karamchedu Mandal**, located in **Bapatla District, Andhra Pradesh**, which was severely affected by **Cyclone Montha on 28 October 2025**. The analysis integrates **Sentinel-1 SAR data** for flood detection and **Sentinel-2 optical data** for vegetation and water index analysis.

The primary objectives of this study are:

- To map flood inundation using SAR backscatter changes,
- To identify excess surface water using NDWI,
- To assess agricultural damage using NDVI change analysis.

2. Study Area

Karamchedu Mandal is situated in Bapatla District, Andhra Pradesh, a region characterized by **intensive agricultural activity**, low-lying terrain, and proximity to coastal systems. The area is highly vulnerable to cyclonic rainfall and flooding during the northeast monsoon season.

The mandal predominantly comprises paddy fields and irrigated croplands, making it particularly sensitive to prolonged waterlogging and inundation events.

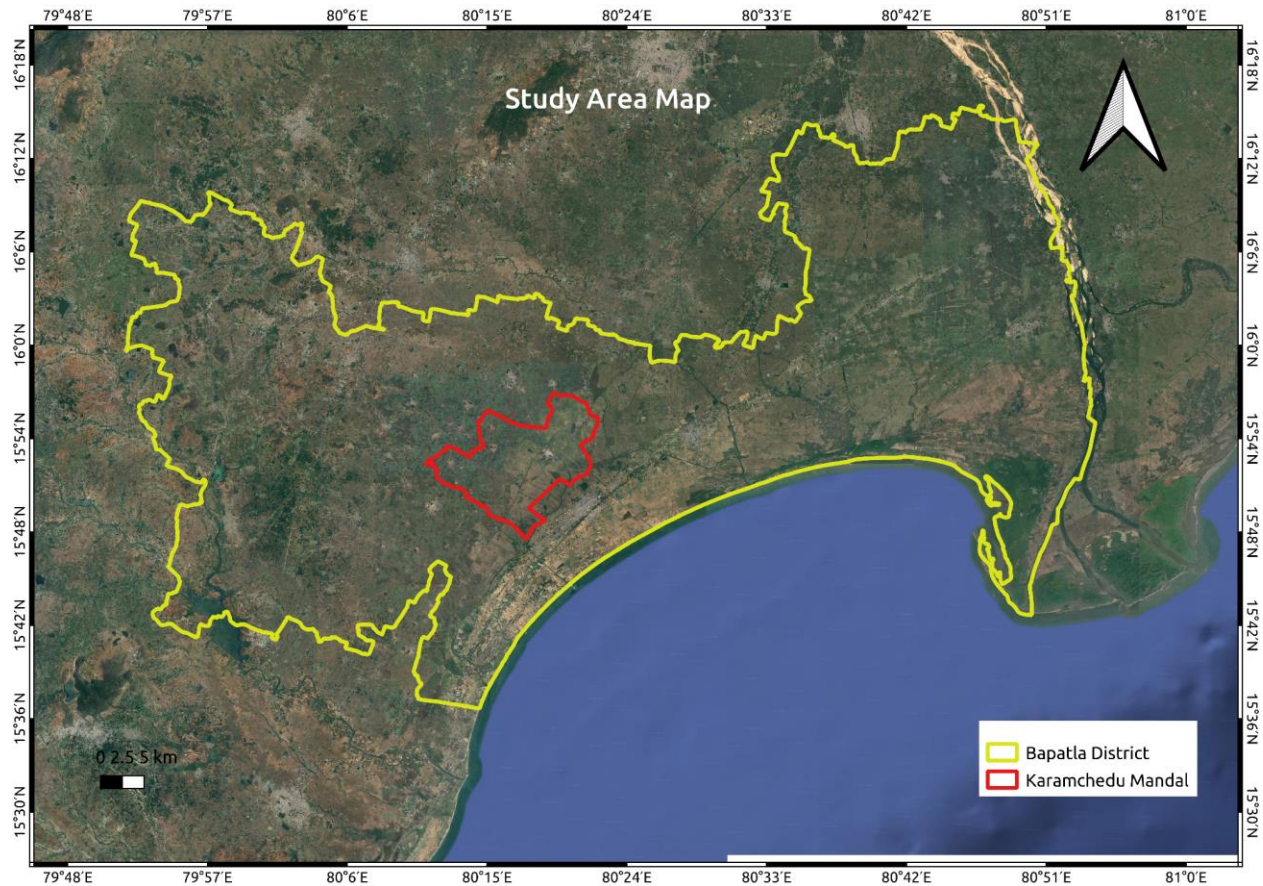


Figure 1 Study Area Map

3. Datasets Used

3.1 Sentinel-1 SAR Data

The first set of data comes from Sentinel-1 equipped with Synthetic Aperture Radar (SAR) instruments. Unlike optical cameras that are sensitive to cloud cover, SAR technology offers exceptional detail, providing a resolution of 10 meters for the Inundation Mapping.

- Product Type: GRD (Ground Range Detected)
- Imaging Mode: IW (Interferometric Wide Swath)
- Polarizations: VV and VH
- Spatial Resolution: 10 m
- Data Source: ASF Search API

Two SAR acquisitions were used:

- **Pre-event:** Late October 2025 (before Cyclone)
- **Post-event:** Early November 2025 (after cyclone)

3.2 Sentinel-2 Optical Data

The second set of data is acquired from the Sentinel-2 satellite, which is equipped with a multispectral optical imaging instrument. Sentinel-2 captures reflected solar radiation across visible, near-infrared, and shortwave infrared bands, enabling detailed analysis of land surface features. With a spatial resolution of up to 10 meters in key spectral bands, Sentinel-2 data is highly effective for land cover classification, vegetation monitoring, and surface water delineation under cloud-free conditions.

- Spatial Resolution: 10 m
- Data Source: Copernicus Open Access Hub

Sentinel-2 imagery was used to compute:

- **Normalized Difference Vegetation Index (NDVI)** for agricultural impact assessment,
- **Normalized Difference Water Index (NDWI)** for identifying excess surface water.

4. Methodology

The methodology adopted in this study integrates **Synthetic Aperture Radar (SAR)** and **optical multispectral data** to map flood inundation and assess its impact on agricultural areas. The overall workflow, as illustrated in Figure 2, consists of data preprocessing, index generation, change analysis, and result interpretation. This multi-sensor approach ensures reliable flood detection under adverse weather conditions and enables comprehensive impact assessment. By combining the all-weather imaging capability of SAR with the spectral sensitivity of optical data, this multi-sensor approach ensures **reliable flood detection even under persistent cloud cover**. Additionally, integration enables a comprehensive assessment of both **inundation extent and associated vegetation stress**, providing a more complete understanding of flood impacts on the landscape.

Methodology Flowchart

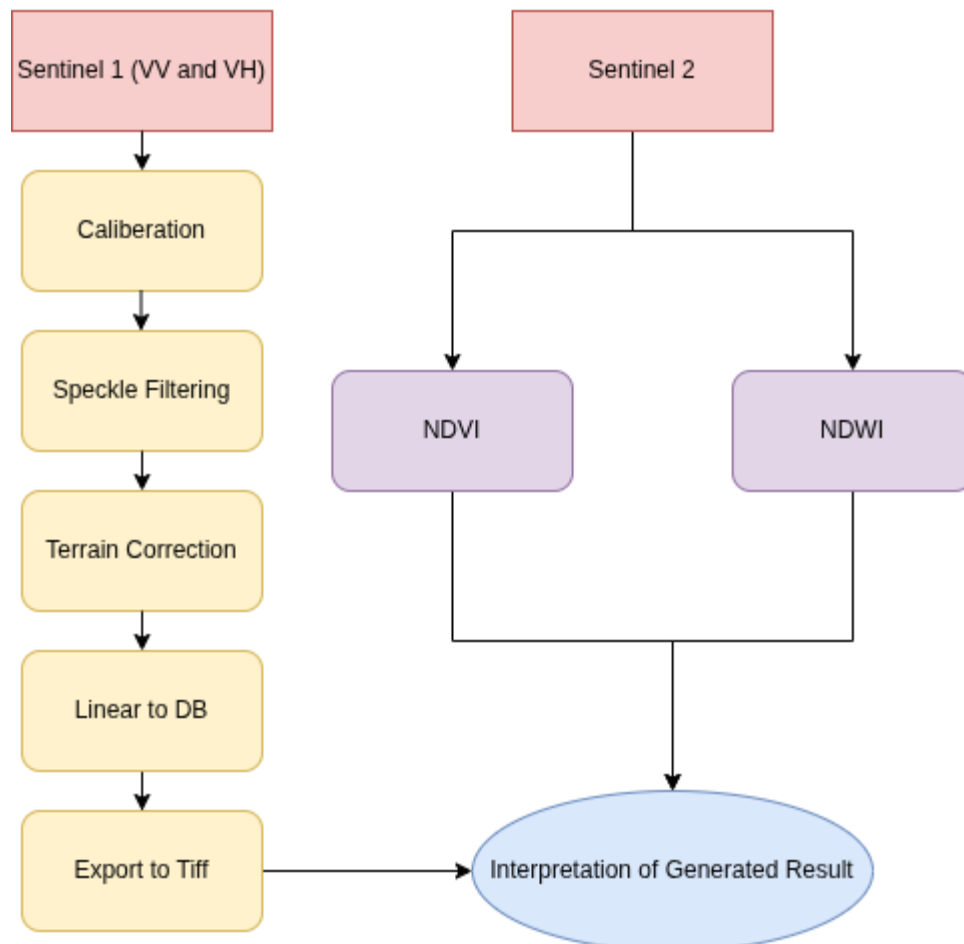


Figure 2 Methodology Flowchart

4.1 SAR Preprocessing using SNAP

Sentinel 1C data, requires a series of processing steps before it becomes a clear and interpretable image. This processing begins by doing **Radiometric calibration**. This step addresses variations in the sensor's response to the incoming radar signal. The data might have uneven brightness or intensity across the image, making it difficult to distinguish features based on their backscatter strength. Radiometric calibration adjusts the brightness and contrast on a photo, ensuring features appear with their true intensity. **Speckle filtering** solves a major challenge in SAR imagery – speckle. This grainy noise arises from the interaction of the radar signal with Earth's rough surface. Speckle filtering techniques aim to remove this noise while preserving the actual image information. **Terrain correction** addresses distortions caused by Earth's varying topography. Without correction, features on slopes might appear elongated or

displaced. Terrain correction takes into account the elevation data to "flatten" the image and represent the true positions of objects. This is crucial for accurate measurements of distances or areas within the image.

4.2 Flood Inundation Mapping Using SAR

Flooded areas appear as **low backscatter regions in SAR imagery** due to specular reflection from smooth water surfaces. Flood inundation was identified using **Backscatter difference analysis** between post- and pre-event images,

4.3 NDWI – Excess Surface Water Detection

The **Normalized Difference Water Index (NDWI)** was generated from Sentinel-2 imagery using:

$$NDWI = \frac{(NIR - SWIR)}{(NIR + SWIR)}$$

NDWI highlights open water and saturated surfaces. Post-cyclone NDWI maps clearly indicate regions with excess water accumulation, complementing SAR-based flood detection.

4.4 NDVI – Agricultural Impact Assessment

The **Normalized Difference Vegetation Index (NDVI)** was computed to assess vegetation health before and after the cyclone:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

A decrease in NDVI values in the post-event image indicates:

- Crop submergence,
- Waterlogging stress,
- Physical damage to standing crops.

NDVI difference analysis was used to quantify the extent of agricultural impact due to inundation.

5. Results and Observations

5.1 Flood Inundation Extent

SAR-based flood maps show extensive inundation across low-lying agricultural fields in Karamchedu Mandal following Cyclone Montha. The backscatter difference image effectively highlights newly inundated areas, even in regions obscured by cloud cover in optical imagery.

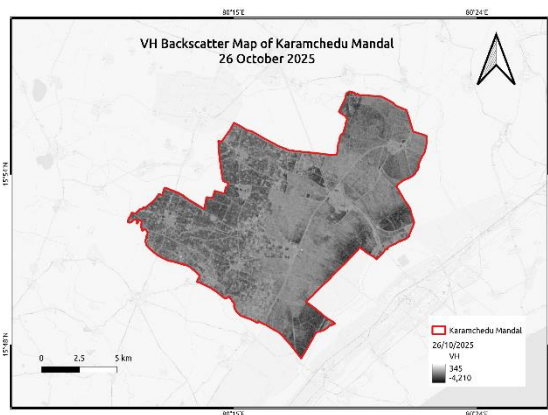


Figure 3 Pre Event VH Backscatter Image

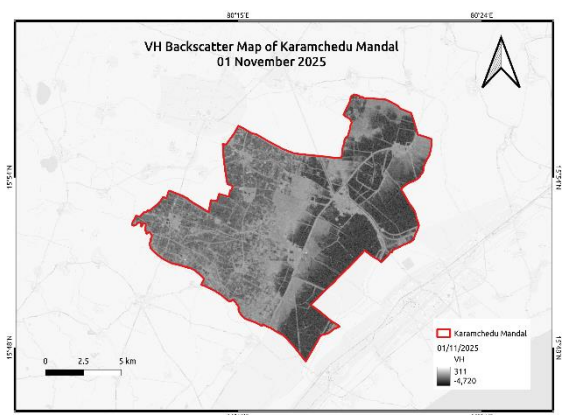


Figure 4 Post Event VH Backscatter Image

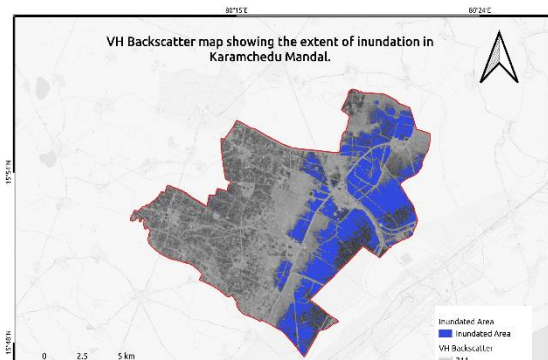


Figure 5 Backscatter Map showing extent of Inundation

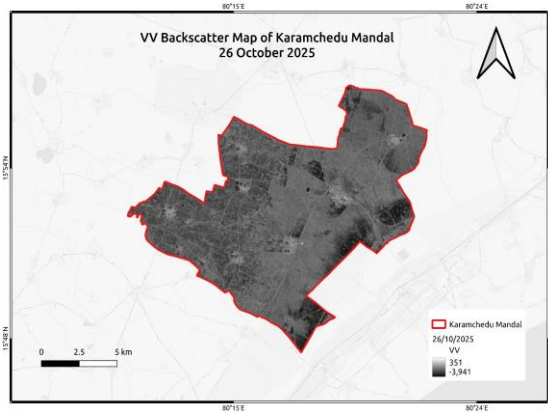


Figure 6 Pre Event VV Backscatter Image

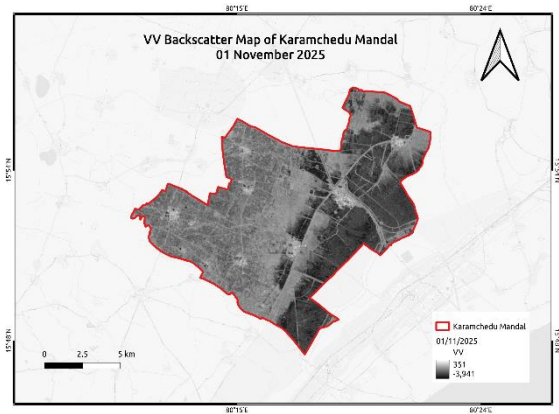


Figure 7 Post Event VH Backscatter Image

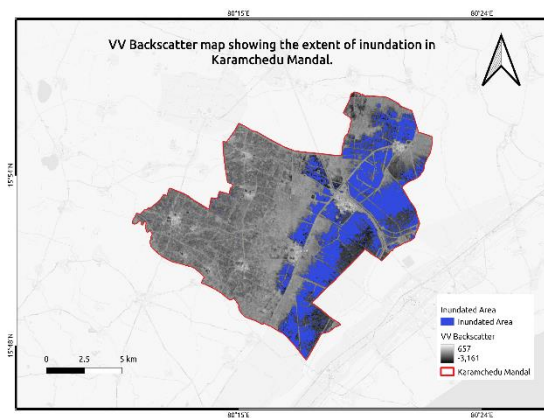


Figure 8 Backscatter Map showing extent of Inundation

5.2 Excess Water Distribution (NDWI)

NDWI results reveal widespread surface water accumulation post-cyclone, particularly in agricultural plots and drainage channels. High NDWI values spatially coincide with SAR-detected flood regions, validating the inundation mapping.

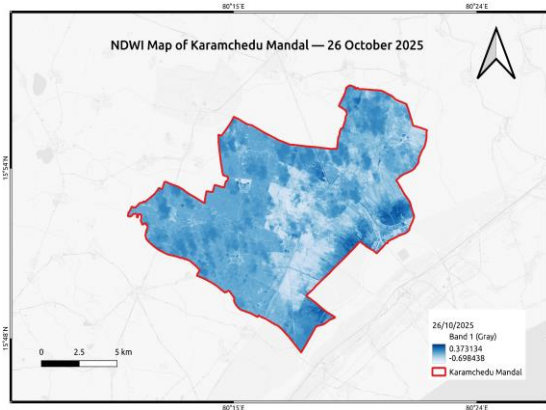


Figure 9 Pre Event NDWI Map

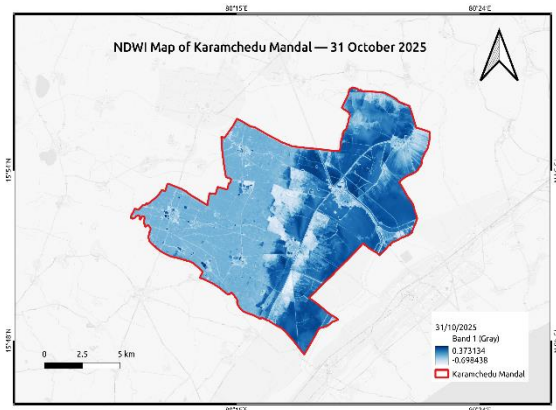


Figure 10 Post Event NDWI Map

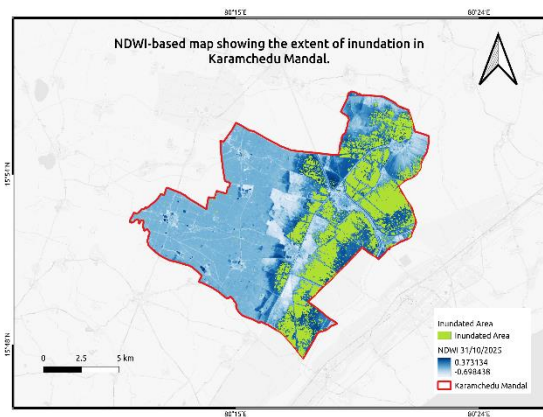


Figure 11 NDWI Map showing extent of Inundation

5.3 Agricultural Damage Assessment (NDVI)

NDVI analysis indicates a significant reduction in vegetation vigor across large portions of the mandal. The observed NDVI decline corresponds closely with flooded areas, suggesting extensive crop stress and potential yield loss due to prolonged inundation.

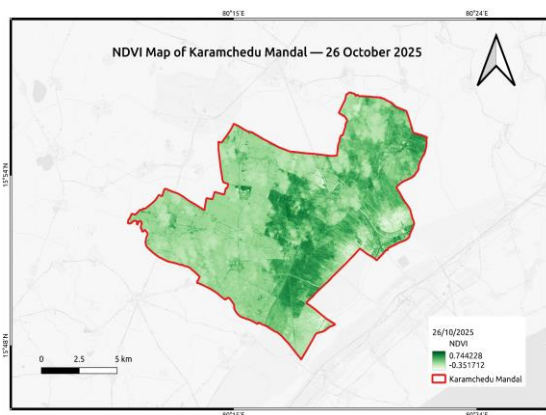


Figure 12 Pre Event NDVI Map

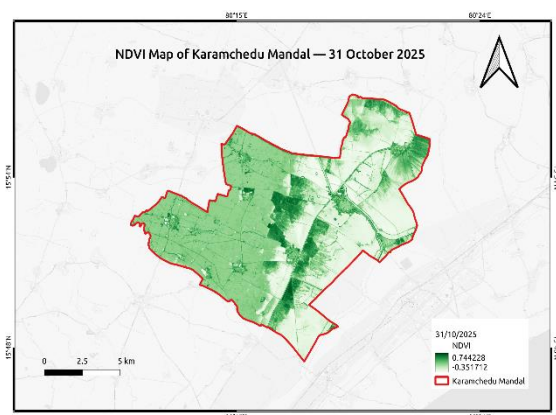


Figure 13 Post Event NDVI Map

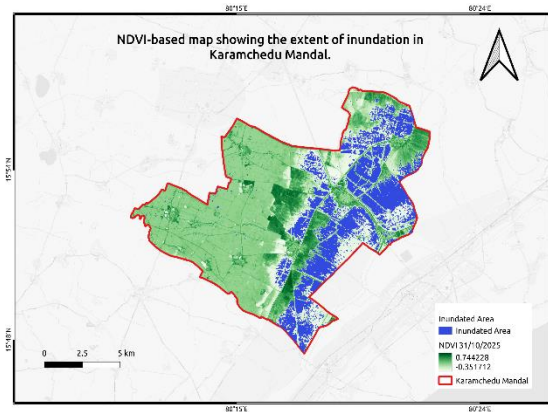


Figure 14 NDVI Map showing extent of Inundation

6. Conclusion

This study demonstrates the effectiveness of an **integrated EO–SAR approach** for rapid flood inundation mapping and agricultural impact assessment following **Cyclone Montha** over **Karamchedu Mandal, Bapatla District, Andhra Pradesh**. The combination of **Sentinel-1 SAR** and **Sentinel-2 optical data** enabled reliable detection of flood-affected areas and associated vegetation stress under challenging post-cyclone conditions.

Sentinel-1 SAR backscatter analysis using both **VV and VH polarizations** successfully identified inundated regions through pronounced reductions in backscatter, highlighting SAR's capability to detect surface water irrespective of cloud cover and illumination conditions. The use of temporal backscatter difference further enhanced the discrimination between flooded and non-flooded areas, ensuring robust flood extent mapping during peak monsoon conditions.

Complementary analysis using **Sentinel-2 derived NDWI** effectively delineated areas of excess surface water, while **NDVI analysis** revealed substantial declines in vegetation health across large agricultural tracts. The spatial agreement between SAR-detected inundation, high NDWI values, and reduced NDVI confirms that prolonged waterlogging significantly impacted agricultural fields within the mandal.